

B4-MASS-GAP evidence-chain migration record (migration batch 3)

TECT verification-first repository · claim B4-MASS-GAP

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1. Purpose and scope

This note is the permanent record of migration batch 3 (plan phase M1, [governance/migration-plan.md](#) §4 priority 1, completing the Sector-B critical path): the evidence chain cited by claim B4-MASS-GAP was copied verbatim from the frozen legacy repository ([TECT2/Contents/Docs/math/](#), [Runs/](#)) into [archive/legacy/](#) and the card’s **legacy**: pointers were resolved to [archive/...](#) paths. This note makes no new mathematical claim and performs no tier action. It is citable as the migration provenance for the B4-MASS-GAP card. The claim’s reproduction status is unchanged (PACKAGE-PENDING): migration relocates and certifies the byte-identity of the textual evidence, not the re-executability of the production-PDE numerics (§3).

2. What was migrated

Seven legacy notes and one run-provenance manifest, covering theory tags Math01 (v2), Math56 (constraint-cone cluster), and Math82 (Addenda G/G2/G3):

Tag	Legacy file (verbatim)	Role
Math01	TECT-Math01-v2-BCC-uniqueness-rigorous.tex.txt	uniqueness within cone
Math56	TECT-Math56-AddB-ClassII-guarded-quotient-analytical.tex.txt	constraint cone (canonical)
Math56	TECT-Math56-Addendum.tex.txt	cone-cluster base
Math56	TECT-Math56-HessJump-audit.tex.txt	cone Hessian-jump audit
Math82	TECT-Math82-Addendum-G-Phase-Z-7point-bifurcation-curve.tex.txt	continuation curve
Math82	TECT-Math82-Addendum-G2-PCG-and-stall-mechanism-audit.tex.txt	stall-mechanism audit
Math82	TECT-Math82-Addendum-G3-vacuum-floor-guard-implementation.tex.txt	vacuum-floor guard
Math82	math82H_groundstate_N32_Lbcc7_MANIFEST.md (artefact)	continuation-run provenance

Disposition: **MIGRATED-VERBATIM** (all eight). Layout: [archive/legacy/{notes/Math01/, notes/Math56/, notes/Math82/, artefacts/Math82/}](#); per-file original paths in [archive/MIGRATION-LEDGER.md](#) (batch 3); per-tag lookup in [archive/legacy/INDEX.md](#). The Math82 *second-order* audit ([TECT-Math82-Addendum-G4-second-order-audit.tex.txt](#)) is available in the legacy corpus but is not cited by the B4 card; it is left for demand-driven migration (§1 pull-based principle).

3. Re-validation status (honest grade)

Three-axis account, distinguishing textual verbatim-fidelity (achieved) from numerical re-execution (waived):

- (a) *Verbatim text fidelity* — **ACHIEVED**. All seven migrated notes are byte-identical to their frozen legacy originals: SHA-256 prefixes `6aeffab9` (Math01-v2), `a5546466` / `26255775` / `4c99f5f9` (Math56 cluster), `9c76de35` / `2025a665` / `fd038f4a` (Math82 G/G2/G3) match source and archive.
- (b) *Provenance cross-check* — **ACHIEVED**. The migrated manifest records continuation point #1 at $\mu^2 = 5.000 \times 10^{-3}$, $r = 2.190 \times 10^{-1}$, converged, with $m^{*2} = +4.247 \times 10^{-2}$ and $\Delta F = +4.150 \times 10^{-10}$ (unfavourable). This reproduces, to every quoted digit, the anchor stated in the **B4-MASS-GAP** card ($m^{*2} = +4.247 \times 10^{-2}$ at $\mu^2 = +5 \times 10^{-3}$, metastable branch). The 7-point table (3 converged, 4 stalled for $\mu^2 \leq -0.5$) is the recorded Regime-III evidence.
- (c) *Numerical re-execution* — **WAIVED (reasoned)**. The anchor is the fixed point of a production Newton–Krylov continuation (`continuation_mu2_v25.py` v2.6.4) seeded by a BCC analytic state (`.npy`, gitignored) on an $N = 32$, $L_{\text{bcc}} = 7$ spectral grid. It is not sandbox-reproducible without the production solver stack and seed. Re-execution is therefore deferred to an operator-side reproduction bundle; the B4 reproduction status stays **PACKAGE-PENDING**, exactly as before migration. Migration does *not* discharge the reproduction obligation; it only relocates and certifies the textual evidence and the run manifest.

4. Convention check

Pillar-1 mass-gap content in the corrected production-kernel convention $r_{\text{braz}} = K(q_0) = \mu^2$. The Math82 continuation is parameterised directly in μ^2 (the manifest’s r column is the affine image $r = \mu^2 + 0.2140336$ of the same lineage), so the migrated curve is in the corrected convention; no stale-convention content. Math01-v2 (BCC uniqueness) and the Math56 cone cluster are structural / variational and convention-neutral.

5. Scope preservation (no tier action)

The migration preserves the card’s pinned scope exactly and upgrades nothing: the result is **CLOSED@SINGLE-MODE-CONE** (T5), the legacy ‘PROVED’ label being precisely this pinned scope. The m^{*2} anchor is the *metastable* subset-4-cosine branch, *not* the ground state; the Regime-III branch interpretation ($\mu^2 \leq -0.5$) remains **undetermined** per the migrated G2 audit. No claim of ground-state uniqueness beyond the single-mode cone is made or implied.

6. Devil’s-advocate (three objections)

- (α) “A migration that waives re-execution certifies nothing.” **VALID** with mitigation: it certifies precisely two things — byte-identity of the textual evidence (§3a, SHA-256) and internal consistency of the card’s quoted anchor with the migrated manifest (§3b) — and makes no stronger claim. The card’s reproduction tier is left at **PACKAGE-PENDING** so the waiver is visible, not hidden. This is the honest disposition for a production-PDE numerical anchor under [governance/migration-plan.md](#) §6 (“re-validation evidence: re-run artefact *or* reasoned waiver”).
- (β) “The anchor is a metastable, unfavourable ($\Delta F > 0$) point — it does not support a mass-gap claim.” **DISMISSED** as a scope objection: B4 claims uniqueness *within the single-mode constraint cone* at the operating point, not global ground-state favourability. The ΔF sign and the metastable label are carried verbatim into §5 and the card; no favourability is asserted. The favourable point in the same table is #3 ($\mu^2 = -0.1$, $\Delta F = -6.27 \times 10^{-10}$), also migrated.

- (γ) “*Archive copies could drift from the frozen legacy originals.*” DISMISSED: the legacy repository is frozen read-only (M0); copies are byte-identical at migration time (§3a hashes recorded), ledgered, and git-tracked from this commit — any later divergence is visible in history.

Quantitative sanity check (mandatory): the manifest arithmetic $r = \mu^2 + 0.2140336$ holds at point #1 ($5.0 \times 10^{-3} + 0.2140336 = 0.2190336 = r$) and at point #4 ($-0.5 + 0.2140336 = -0.2859664 = r$), confirming the $\mu^2 \leftrightarrow r$ affine map across converged and stalled regimes; the seven wall-times ($O(20\text{--}1075\text{s})$) sum to ≈ 45 min, consistent with a single 7-point production sweep, not a multi-day run.

7. Result footer

Result ID:	B4-MASS-GAP (supporting migration record)
Precise statement:	The B4 evidence chain (Math01-v2, Math56 cone cluster, Math82-AddG/G2/G3 + continuation manifest) was migrated verbatim (7/7 SHA-256 match) and the card’s legacy: pointers resolved to archive/ paths.
Scope:	Textual verbatim-fidelity + manifest-provenance consistency only; no numerical re-execution; no new mathematical claim; no tier action.
Dependencies:	archive/legacy/ batch-3 file set (ledger 2026-06-23); A1-KERNEL-CONV (hard dependency, already migrated).
Evidence grade:	ANALYTIC (text) + provenance (manifest); numerical re-execution WAIVED (PACKAGE-PENDING).
Reproduction command:	Production-PDE; not sandbox-reproducible. Operator-side: continuation_mu2_v25.py v2.6.4, N=32, Lbcc=7, BCC analytic seed; deferred reproduction bundle.
Expected output:	Continuation point #1 $m^2 = +4.247\text{e-}02$ at $\mu^2 = +5\text{e-}03$ (metastable branch), reproducing the card anchor.
Falsification gate:	A second distinct minimiser within the single-mode + cone class at the operating point, or failure to reproduce the anchor within stated tolerance.
Tier before / after:	B4 T5 / T5 (no change); migration-clean: no / yes.
No-overclaim statement:	No ground-state uniqueness beyond the single-mode cone; the subset-4-cosine anchor is metastable, not the ground state; Regime-III remains undetermined.
Next required action:	Migrate B3-BCC-STRUCT (batch-3 continuation), then operator-side reproduction bundle for the B4 anchor.